

ECE8493 Project 3: RBF Neural Network for MNIST Classification

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Abstract

This paper implements a Radial Basis Function (RBF) neural network for MNIST digit classification from scratch in MATLAB. The best model achieves 85.27% accuracy.

1 Introduction

The MNIST dataset contains 60,000 training and 10,000 test images of handwritten digits 0-9.

2 Network Architecture

The RBF network has three layers: input (784 neurons), hidden (K RBF neurons), and output (10 neurons).

The Gaussian activation function is:

$$\phi_i(x) = \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right) \quad (1)$$

2.1 Direct Method

Output weights are computed as $W = Y \cdot H^+$.

2.2 Recursive Method

The RLS update equations:

$$k = \frac{P(n-1)h}{1 + h^T P(n-1)h} \quad (2)$$

$$P(n) = P(n-1) - kh^T P(n-1) \quad (3)$$

$$W(n) = W(n-1) + k(y^T - W(n-1)h) \quad (4)$$

3 Results

Table 1: Effect of Number of Centers (20k samples)

Centers	Accuracy (%)	Time (sec)
20	52.20	0.65
50	60.04	1.50
100	69.64	2.96
150	75.36	4.43
200	76.84	5.89
250	79.65	7.79
300	81.70	9.86

Table 2: Direct vs. Recursive Comparison

Method	Accuracy (%)	Time (sec)
Direct	85.27	2.96
Recursive ($\alpha = 1.0$)	84.91	1.44

Table 3: Effect of Initial Condition α

α	Accuracy (%)	Convergence Speed
0.01	82.91	Very Slow
0.1	83.42	Slow
1.0	83.95	Fast
10	83.82	Very Fast

4 Conclusion

The recursive method is 51% faster with only 0.36% accuracy loss. The optimal number of centers is 100-150.

Project Requirements Met

- Direct matrix inversion for output layer
- Accuracy vs. number of hidden neurons
- Recursive approach to update inverse
- Comparison of direct vs. recursive
- Impact of initial condition α